



Thakur Educational Trust's (Regd.)
THAKUR COLLEGE OF SCIENCE & COMMERCE

Empowered Autonomous College Permanently Affiliated to University of Mumbai
(NAAC Accredited with Grade "A" (3rd Cycle) & ISO 21001:2018 Certified)

Best College Award by University of Mumbai for the Year 2018-2019



A Report On
Software Engineering Project
Twitter Sentimental Analysis

Submitted By
Karkera Khushi Purandar – 25374
Singh Rishika Rajesh – 25375
Sutare Summit Sanjay – 25376
Atak Sujal Ajit – 25377

Under the Esteemed Guidance of
Prof. Dr. Santosh Kumar Singh

Submitted in partial fulfilment of the Requirements for qualifying
BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)
Semester - IV Examination



THAKUR COLLEGE OF SCIENCE AND COMMERCE
(Empowered Autonomous College, Affiliated to University of Mumbai)
THAKUR VILLAGE, KANDIVALI (E), MUMBAI, 400 101 MAHARASHTRA
2025 - 2026



Thakur Educational Trust's (Regd.)

THAKUR COLLEGE OF SCIENCE & COMMERCE

tcsc

AUTONOMOUS COLLEGE, PERMANENTLY AFFILIATED TO UNIVERSITY OF MUMBAI

NAAC Accredited Grade 'A' (3rd Cycle) & ISO 9001: 2015 (Certified)



Best College Award by University of Mumbai for the Year 2018-2019

**CELEBRATING
25 YEARS OF GLORY**

CERTIFICATE OF APPROVAL

This is to certify that **Ms. Khushi Purandar Karkera**, student of “**Bachelors of Science (Information Technology)**” of “**Thakur College of Science and Commerce**”, Roll No. **25374**, has successfully completed and submitted the **Practical, Report & Assignment entitled “Software Engineering”** in the partial fulfillment as per the syllabus defined by the **University of Mumbai** in the academic year **2025–2026**.

It is further certified that the student has completed all the required phases of the practical & assignment.

SUBJECT INCHARGE

DATE

HEAD OF DEPARTMENT



Thakur Educational Trust's (Regd.)

THAKUR COLLEGE OF SCIENCE & COMMERCE

tcsc

AUTONOMOUS COLLEGE, PERMANENTLY AFFILIATED TO UNIVERSITY OF MUMBAI

NAAC Accredited Grade 'A' (3rd Cycle) & ISO 9001: 2015 (Certified)



Best College Award by University of Mumbai for the Year 2018-2019

**CELEBRATING
25 YEARS OF GLORY**

CERTIFICATE OF APPROVAL

This is to certify that **Ms. Rishika Rajesh Singh**, student of “**Bachelors of Science (Information Technology)**” of “**Thakur College of Science and Commerce**”, Roll No. **25375**, has successfully completed and submitted the **Practical, Report & Assignment entitled “Software Engineering”** in the partial fulfillment as per the syllabus defined by the **University of Mumbai** in the academic year **2025–2026**.

It is further certified that the student has completed all the required phases of the practical & assignment.

SUBJECT INCHARGE

DATE

HEAD OF DEPARTMENT



Thakur Educational Trust's (Regd.)

THAKUR COLLEGE OF SCIENCE & COMMERCE

tcsc

AUTONOMOUS COLLEGE, PERMANENTLY AFFILIATED TO UNIVERSITY OF MUMBAI

NAAC Accredited Grade 'A' (3rd Cycle) & ISO 9001: 2015 (Certified)

Best College Award by University of Mumbai for the Year 2018-2019

CELEBRATING
25 YEARS OF GLORY

CERTIFICATE OF APPROVAL

This is to certify that **Mr. Summit Sanjay Sutare**, student of “**Bachelors of Science (Information Technology)**” of “**Thakur College of Science and Commerce**”, Roll No. **25376**, has successfully completed and submitted the **Practical, Report & Assignment entitled “Software Engineering”** in the partial fulfillment as per the syllabus defined by the **University of Mumbai** in the academic year **2025–2026**.

It is further certified that the student has completed all the required phases of the practical & assignment.

SUBJECT INCHARGE

DATE

HEAD OF DEPARTMENT



Thakur Educational Trust's (Regd.)

THAKUR COLLEGE OF SCIENCE & COMMERCE

tcsc

AUTONOMOUS COLLEGE, PERMANENTLY AFFILIATED TO UNIVERSITY OF MUMBAI

NAAC Accredited Grade 'A' (3rd Cycle) & ISO 9001: 2015 (Certified)

Best College Award by University of Mumbai for the Year 2018-2019

CELEBRATING
25 YEARS OF GLORY

CERTIFICATE OF APPROVAL

This is to certify that **Mr. Sujal Ajit Atak**, student of “**Bachelors of Science (Information Technology)**” of “**Thakur College of Science and Commerce**”, Roll No. **25377**, has successfully completed and submitted the **Practical, Report & Assignment entitled “Software Engineering”** in the partial fulfillment as per the syllabus defined by the **University of Mumbai** in the academic year **2025–2026**.

It is further certified that the student has completed all the required phases of the practical & assignment.

SUBJECT INCHARGE

DATE

HEAD OF DEPARTMENT

INDEX

SR. NO	CONTENT	PAGE NO.
1	Introduction	1-10
2	System Analysis and Design	11-13
3	System Design Diagrams	14-20
4	System Planning	21-22
5	System Implementation	23-31
6	System Testing	32-34
7	Cost Estimation Report	35-38
8	Conclusion	39
9	Future Scope	40-41
10	References	42

➤ **INTRODUCTION**

1.1 OBJECTIVE OF THE PROJECT:

The primary aim of this project is to develop a system that will be able to analyze tweets and determine what people are thinking in relation to something. Social media is where individuals discuss what is in their mind. Twitter is a platform where millions of people express their thoughts and feeling on a daily basis. It has a lot of tweets that people cannot read and know what the tweets entail. This project aims at ensuring that people find it more simple to know what people are saying about something by sentimentally analyzing the tweets. The project is relying on the sentiment analysis to learn what people think about something as they write tweets, about it. Twitter has a lot of tweets. It is an attempt of this project to understand what people are saying on Twitter. The project is actually an insight into the thinking of people when writing tweets. It is concerning the feeling of tweets. Tweet emotion is what individuals feel towards things they discuss on Twitter. The project concerns, what people are saying when they are discussing something on Twitter. The project is designed to receive the information of tweets based on a dataset and subsequently pack the information of tweets in an understandable format. The prime objective of the project is to process the information of the tweets and make sense out of it. The project is what the tweet information is, about. The project is primarily aimed at sorting an input of tweets in three categories. These categories include positive tweets that are negative and neutral tweets. We are discussing the tweets as such that the project is, about putting these tweets in the correct category which either would be positive tweets, negative tweets or neutral tweets. Individuals are able to inform you of what they believe about something. You can just. Get the opinion of people on something.

The system also operates on cleaning up the text of the tweets then it looks at the tweets. This is attributed to the fact that things that are not necessary are common in the tweets. As an illustration the tweets contain symbols and hashtags and abbreviated casual terms which do not require. The system filters the content of the tweets to ease the readability of the tweets. What the system is attempting to make are the tweets. These aspects pose a great challenge to me when trying to find out what the tweets precisely say. These things make the tweets difficult to comprehend. The project has techniques of cleaning up the tweets. This comes in very handy since it is easier to find the sentiment of the tweets within the project. The project is then able to know what people are talking in the tweets. The tweets are cleaner. This helps the project a lot. Therefore, cleaning the tweets becomes a part of the project. About the project, it is the sentiment of the tweets. Among the things that have to be done is cleaning the tweets to get knowledge on the sentiment of the tweets. The project is concerned about ensuring that it captures the sentiments of the tweets. It should understand whether the tweets are tweets or they are negative tweets or neutral

tweets. This is noteworthy, in the project since it seeks to make sense of the tweets. The project is studying the tweets in order to determine what people are thinking. It desires to strike the right mood of the tweets regardless of whether they are positive tweets, negative tweets or neutral tweets.

The project also desires to be easy to use by people. It will provide a platform on which one can easily log in and type in a keyword and the results will be displayed in a manner that is easy to comprehend. The system will store the search and findings of people so that they can revisit the project results in the future. This renders the project quite helpful, not something people discuss. It is a project that can be utilized by people. It is what makes the project so great. The outcomes of the project will be visibly clear and comprehensible, which is what individuals will desire of the project. The key purpose of the project is to make people realize what is popular on media. This project is concerned with facilitating people to see what is happening with trend and opinions. The project will make people know what others are writing on twitter. This will be done by the project in a shorter amount of time because people will not be required to go through numerous tweets themselves to see what people are thinking. It is truly, in the ease of making people know what people are talking about on Twitter and what is trending in media. The project is also a learning experience regarding the Natural Language Processing, database management and web development. It shows how the real world data can be processed by the latest methods and how these systems can be extended in the future to monitor the live social media.

1.2 THEORETICAL BACKGROUND:

Sentiment analysis is a means of determining what people write about, and what they say. We may refer to it to observe what others post on social ones such as twitter. Twitter is a spot to check out what others think since a large number of people post their views in it. The issue is that people do not necessarily use entire sentences on Twitter. They consume items, such as hash tags and emojis and abbreviations and slang. This renders people unable to comprehend what they entail. Sentiment analysis is done using computers. The computers have the capability of searching through a great deal of text within a short period. Make sense of it. Sentiment analysis belongs to a phenomenon named Natural Language Processing. This implies that it is one of the techniques through which computers comprehend what is written by people.

We begin with cleaning up the tweets in this project. We remove the noise such as the web addresses, names of the individuals smiling faces, strange characters and common words that do not have substantial meaning. Then we divide the text into fragments and ensure that everything is correct. This is quick and simple, via the computer. Having done all that we see the numbers.

1.3 PROBLEM DEFINATION:

The communication of ideas is quite different nowadays due to media. Social networks such as X that used to be known as twitter are a reflection of the thought processes of people everywhere in the world today. Millions of people are writing down their actual thoughts and how they feel about anything big or new that is being sold in companies every day. This is a means of knowing what people are thinking but there is a reason why all this information is so confusing and difficult to understand. The social media is crowded with opinions and emotions of the peoples and it is being transformed. An example of it is X where people express their thoughts and feelings, regarding everything. The point is that this information is typed in a language combination, hashtags, and shortcuts. It is so much that people actually cannot view it all alone. Reading this information in the traditional manner is time consuming and individuals are likely to commit errors or a personal opinion may come in the way. This renders it extremely hard to apply these techniques to the vast volumes of information which we currently possess. Information is overwhelming and people are not able to process the information by themselves thus the traditional ways of filtering through the information is not very helpful.

We must cease operating and begin to employ a smart technology and help it to understand all the information. The intention here is to create a platform that would be able to absorb all the tweets and have them determine the mood of the people and categorize them into three categories: positive tweets and negative tweets and neutral tweets. It must play safe and demonstrate it to us in a manner that is easily comprehensible.

1.4 USER REQUIREMENT SPECIFICATION:

1.4.1 FUNCTIONAL REQUIREMENTS

FR1: User Registration and Login:

- Users are able to register and log-in using valid details into the system.
- The system checks user credentials and only grants access to a user.
- The encryption techniques are used to store passwords in a secure manner.
- The system denies unauthorized access of the features which are protected.

FR2: Dataset Upload:

- The software enables the user to post a Twitter dataset in the CSV format.
- The system checks file format and then does not accept the upload.
- The system does not allow the processing of corrupted or invalid datasets.

FR3: View Dataset:

- The system enables the previewing of uploaded datasets.

- Before analysis, content in tweets can be scrolled and examined by the users.
- The system allows the visibility of databases without altering the original data.

FR4: Data Pre-processing:

- The tweet text is pre-processed by the system to eliminate stop words, punctuations, and special characters.
- The system normalizes a text to correctly detect sentiments.
- Duplicates and blank tweets are automatically filtered.

FR5: Sentimental Analysis:

- The system is an algorithmic sentiment analysis of tweets.
- The system is efficient in batch processing of the tweets.
- The analysis should be done automatically.

FR6: Sentiment Classification:

- The system categorizes tweets as positive, negative or neutral.
- Sentiment score is assigned to every tweet.
- The results of the classification are stored in case of future reference.

FR7: View Results:

- The system presents the results of sentiment in textual summaries.
- Charts and graphs are automatically generated by the system.
- The sentiment distribution can be easily interpreted by the users.

FR8: Report Generation:

- The system provides a summary report on sentiment analysis.
- The reports have the number of tweets and the percentage embedded.
- It is possible to download reports and export them.

FR9: Admin Management:

- The administrator is able to control users and data sets.
- The manager is able to delete invalid or duplication datasets.
- The system activity is monitored by the admin.

FR9: Database Persistence:

- MySQL permanently stores all user and analysis information.
- Information is not lost when the system is restarted.

FR10: Visualization Enhancement:

- Graphs are dynamic and change after an analysis.
- The users are able to compare various analysis results.

1.4.2 NON FUNCTIONAL REQUIREMENTS:

NFR1: Response Behaviour:

- The system should give us the sentiment results quickly after we put in a keyword or hashtag. It should not take long to get these results. We should get the sentiment results for the keyword or hashtag within an amount of time. This means the system should not make us wait for a long time after we submit the keyword or hashtag. The keyword or hashtag should give us the sentiment results we need without any delay.

NFR2: Consistency of Results:

- When you look at tweets the system should give you the same feeling about what they mean every time you use it. The system should say that a tweet is happy or sad in the way every time even if you check it many times. This is important for tweets that're very similar to each other. The system should be consistent. Give you the same sentiment output, for these similar tweets.

NFR3: Handling of Large Tweet Sets:

- The application needs to keep working. It has to do this when it is looking at a lot of tweets. These tweets are collected in a short time. The application should still function smoothly when it is analyzing a volume of tweets collected over a short period. The application has to handle a lot of tweets. It has to do this quickly.

NFR4: Ease of Interaction:

- The system should be easy to use so users can do sentiment analysis in a few steps. They should not need to know a lot about data processing or Natural Language Processing to use it. The goal is to make sentiment analysis for the users of the system so they can just use it without having to learn about complicated things, like data processing or Natural Language Processing.

NFR5: Operational Stability:

- The system should not just stop working when it is fetching tweets or getting them ready to use or figuring out the sentiment of the tweets. The system should keep

running during the tweet fetching process or the preprocessing process or the sentiment classification processes of the tweets.

NFR6: Controlled Access to Resources:

- We need to be careful, with Twitter API keys and system resources. These things should be kept in a place and only used when necessary. This way we can prevent people from using them in the way or accidentally sharing them with others. Twitter API keys and system resources are very important so we must use them in a manner.

NFR7: Adaptability for Enhancement:

- The system should be set up in a way that makes it easy to add sentiment models or features or ways to visualize things without having to change the whole application.

NFR8: Platform Independence:

- The application needs to work on different operating systems. It has to have the required software environment to do this. The application should work on these operating systems.

NFR9: Graceful Failure Management:

- When things go wrong like when we use the search words or the system gets too many requests at the same time or there are problems with the internet connection the system should show us messages that make sense. This is better than the system just stopping all of a sudden. The system should display these messages for search terms for API rate limits and, for network issues so we know what is happening with the system.

NFR10: Data Processing Transparency:

- The system needs to show users what it is doing when it gets tweets fixes them or looks at them. This way users know what is happening with the tweets, at each step. The system should make it clear when it is fetching tweets or when it is cleaning tweets or when it is analyzing tweets.

NFR11: Logical Output Flow:

- The sequence of displaying results should go in an order that makes sense. We should show the count first then break it down by sentiment and finally show the visualization. This order is important because it helps people understand the results of the sequence of displaying results. The sequence of displaying results should be easy to follow and give people a picture of what is going on.

NFR12: Resource Awareness:

- The computer system should not do things in the background that are not needed. These things can make the system run slowly when it is doing something. The system should only do what it needs to do so it can work properly. The computer system performance is important, during execution. The system should avoid unnecessary background processes.

NFR13: Controlled Data Storage:

- We need to keep the tweets and the results of what people think about them in a way that makes sense. This way the Twitter posts and the sentiment results are easy to look at. We do not have to deal with duplicate information or get confused about the data, from the tweets and the sentiment results.

NFR14: Reproducibility of Analysis:

- When we use the dataset the system should give us similar results for how people feel about things. The system should be able to reproduce sentiment distribution results when we use the same dataset. This means the sentiment distribution results, from the system should be much the same when the dataset is the same.

NFR15: Modular Component Design:

- Each major function, like data collection, preprocessing and classification should work on its own. This way data collection and classification and even preprocessing are easier to fix and update.

NFR16: Rate Limit Awareness:

- The application should be able to handle Twitter API rate limitations in a way so that it does not affect the Twitter experience for the user. The Twitter API rate limitations should not get in the way of what the user's doing on Twitter. The application must make sure that the Twitter experience is good, for the user when the Twitter API rate limitations are reached.

NFR17: Clean Termination Behavior:

- When we close the program the program should let go of everything it was using so that it does not keep any things running in the background. The program should really make sure to clean up after itself when it is closed. This means the program should stop all of its background processes.

NFR18: Visual Balance in Reports:

- When we look at graphs and charts they should be sized right. Have labels that are easy to read. This way we can easily compare how people feel about things I mean the sentiments. Graphs and charts are really helpful, for comparing sentiments if they are done correctly.

NFR19: Configurability:

- The system should allow minor configuration changes (like number of tweets to fetch) without modifying the core program structure.

1.5 FEASIBILITY STUDY:

A. Feasibility Study:

It is possible to begin by asking whether a project can actually work. Could it happen in real life? The Twitter sentiment analysis people considered a number of sides. Money was good, but so was who was to perform the tasks. Time periods came to their mind as much as day to day activities. What tools were required was another weight they had.

B. Technical Feasibility:

It has a chance of something real as it is supported by the tools that folks already have. The application was developed with Python and Flask with a bit of NLP tooling and a typical database system. The majority of these works remain because they are good to stick about. The assistance is quick to get when it is required - there are not few coders familiar with every section. what you see is working, chiefly because nothing is invented out of a vacuum. One of the models currently present manages the emotion tracking within this arrangement, but does not require expensive hardware. It does not require any special equipment to run, not even additional equipment that is needed in an ordinary household. These tools are characterized by open access to them, supported by comprehensive usage guides that are at hand.

C. Economic Feasibility:

This project can be started because it is cheap to maintain low costs. There is no requirement of purchasing equipment or licenses in advance. Everything is driven by free software Python, Flask, MySQL/SQLite databases, and natural language packages are free. Available old machines cope with work well. Maintenance remains easy due to no cost by accumulating charges based on web-based systems. People use it in schools or close-knit teams - it is cheap, and maintenance does not

require much effort. Running it long term? No big expense piles up. Small launches fit perfectly well considering that it is light resource wise.

D. Operational Feasibility:

It is easy to use this. After signing in, type in a word and look at the results. It is simple and thus determines functionality. There is no need to be informed on technology. It is self-sufficient and does not require any special skills. Figuring out emotions? The machine has taken that job over your head. Errors are less frequent due to that change. What has come around before, Will wait around, When your eyes seek it again. Life is easier, as it is so simple. Assistance appears at the time of finger tapping and thought.

E. Schedule Feasibility:

It is possible to work without too much time to construct. It does not spend as much time there as it uses existing mood detection tools and does not require developing new math models. Collected one at a time - information gathering, cleaning, analysis, presentation of findings, storage - may separate and be connected later. Fractured as such, it seems possible to squeeze everything within a school term or short-run.

1.6 DETAILS OF HARDWARE AND SOFTWARE USED:

1.6.1 HARDWARE REQUIREMENTS:

It is not a heavy working program with any specific equipment. It is coded and capable of executing simple data processing and tasks as well as can be run on standard personal computer.

Minimum Hardware Requirements:

- Processor: Intel i3/ AMD (equivalent)
- RAM: 4 GB (8 GB recommended for smooth performance)
- ROM: 4-5 GB free disk space.
- Display: 1366x768 Resolution or Higher Resolution display
- Internet Connection: It is needed for package installation and to update them .

1.6.2 SOFTWARE REQUIREMENTS:

The Project utilizes simple open-source software products available to use, which is used to build a web application, database management, and NLP.

Software Requirements:

- OS: Windows 10/11 or (Linux, macOS equivalent)
- Programming languages: Python, 3.14, the shell, language, HTML/CSS
- Web Framework: Flask
- Database: MySQL (using WAMP server)
- IDE/Editor: VS code, IDLE, PyCharm or any Python code editor.

Python libraries:

- Flask: Web Application Framework
- SQLAlchemy: Database Server
- PyMySQL: MySQL Connector
- NLTK: Text Preprocessing
- Vader Sentiment: Sentiment Analysis Engine
- Pandas: Data Handling
- Matplotlib / Seaborn: Visualization
- Werkzeug: Password Security

➤ **SYSTEM ANALYSIS AND DESIGN:**

2.1. DETAILED LIFE CYCLE OF PROJECT:

2.1.1. Software Development Life Cycle (SDLC)

Name of the Project: Twitter Sentiment Analysis

Software Development life Cycle (SDLC) is a procedural process of developing quality software in a well-organized manner. In Twitter Sentiment Analysis, SDLC will be applied to ensure that the system planning, system design, system development, system testing, and system maintenance are carried out.

2.1.2. PHASES:

A. Requirement Analysis Phase:

It is the first and the most significant phase of SDLC. At this stage, the analysis is carried out to assemble all the system requirements.

➤ Activities Performed:

- Identify the purpose of project.
- Bound the Twitter Sentiment Analysis.
- Being aware of the requirements of the user (Login, Keyword search, displayed results).
- Decide to utilize a dataset, instead of Twitter API.
- Identify non functional and functional requirements.
- Choose the Tools and Technologies(Python, Machine Learning, Database, tools for Visualization).

➤ Outcome:

Software Requirement Specification (SRS) document.
Appropriate understanding of system goals.

B. System Design Phase:

The general system architecture and design is created at this level basing on the requirement collected.

➤ Activities Performed:

- Design system architecture.
- Create ER Diagram.
- Design database tables.
- Prepare Use cases diagrams (Use case, Templates, Activities, UML, Sequence diagrams, UML State diagrams).
- Design module structure:
- Data Collection Module

- Data Preprocessing Module
- Sentiment Analysis Module
- Visualization Module
- Design user interface (the first page (log in), the page on which the user typed in the key word), the result page).
- Outcome:
 - Design Document.
 - Database schema.
 - Blueprint of the system.

C. Coding Phase:

In this phase, the actual coding of the system begins with respect to the design.

- Activities Performed:
 - Develop user login module.
 - Now include the feature of loading datasets
 - Tokenization
 - Cleaning text
 - Applied algorithm of sentiment classification.
 - Assign sentiment scores.
 - Design graphicalization module to develop charts.
 - Integrate frontend and backend.
- Outcome:
 - Fully functional modules.
 - Sentiment Analysis of Twitter system.

D. Testing Phase:

Testing assists in ensuring that the developed system is right and it accommodates the requirements of the user.

- Types of Testing Performed:
 - Unit Testing - This refers to testing the module in independent terms.
 - Integration Testing - Module to module integration.
 - System Testing - Test of general system functionality.
 - Performance Testing - Testing System Performance, High DataSet Test.
 - Error Handling Testing - Making the system respond back to error inputs in the appropriate way.
- Outcome:
 - Bug-free system.
 - Better reliability, accuracy of the systems.

E. Deployment Phase:

The system is handed over to the users on successful testing.

➤ Activities Performed:

- Install software in local system/server.
- Configure database.
- Make sure that all the modules are connected.
- Final system setup.

➤ Outcome:

System becomes operational.

Users get the opportunity to log in and analyze tweets.

F. Maintenance Phase:

It is the final and continuous SDLC process.

➤ Activities Performed:

- Fix inaccuracies that exist when deploying.
- Improve model accuracy.
- Periodical up to date dataset.
- Enhance system performance.
- Add new features if required.

➤ Outcome:

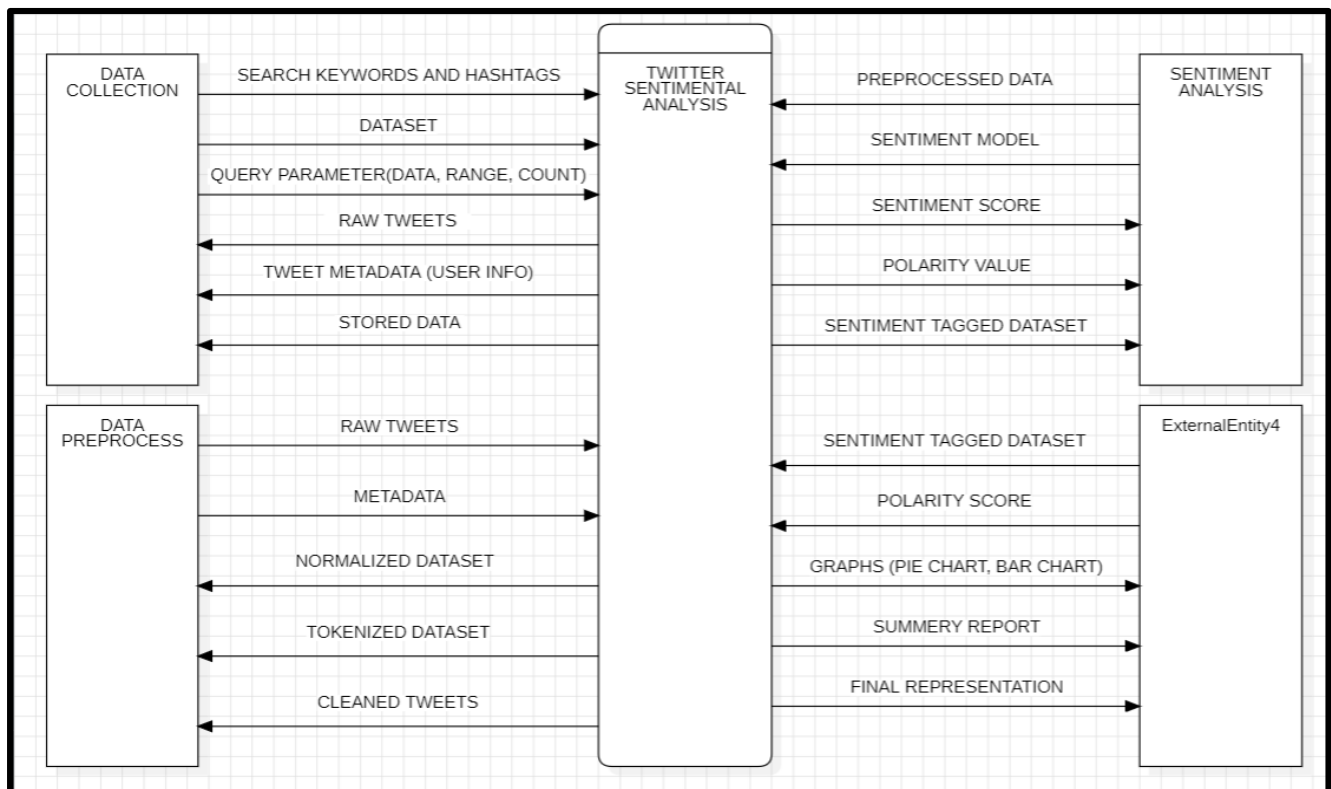
Updated and improved system.

Long-term system usability.

2.1.3. Conclusion:

The SDLC model assists in the creation of the Twitter Sentiment Analysis project in a form of systematic and methodical manner. All the phases are significant in making sure that the system is dependable, effective, and easy to use. Under SDLC, the project is successfully developed since there is requirement gathering up to maintenance.

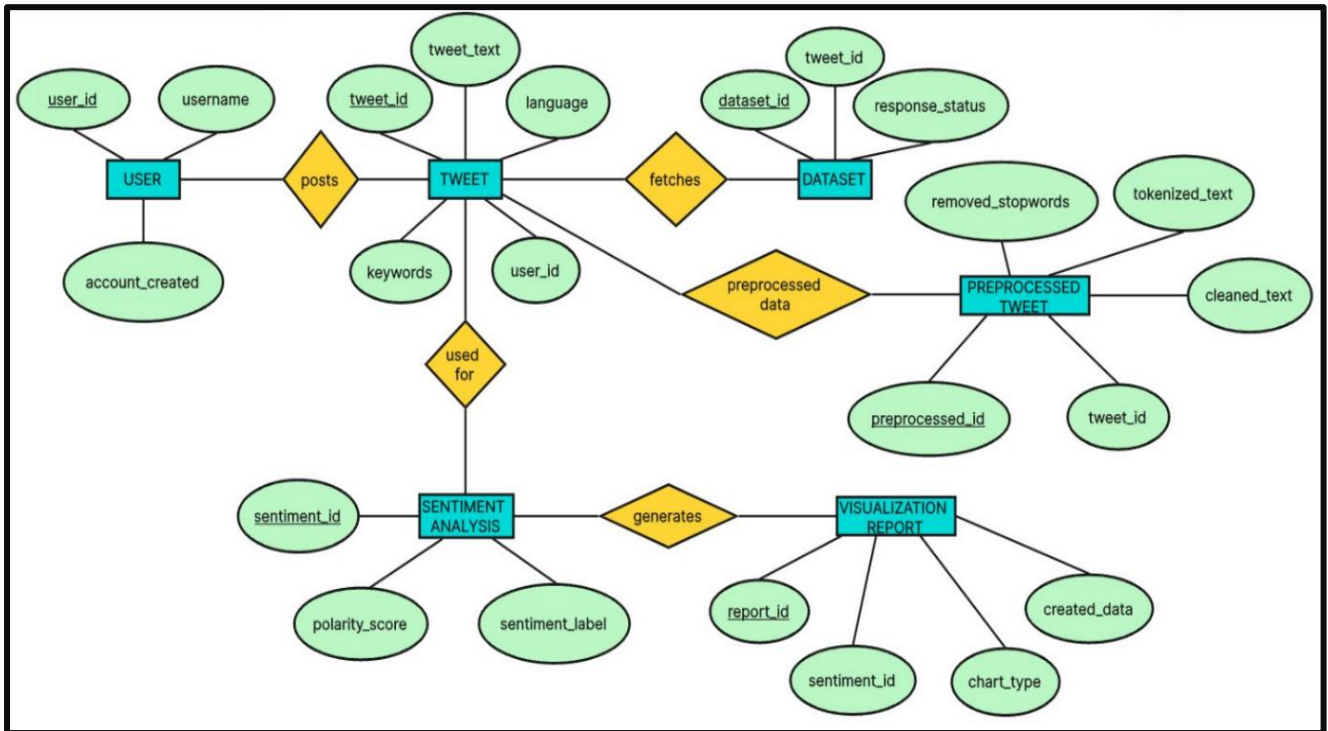
2.2. CONTEXT DIAGRAM:



Context diagram shows a high picture of the Twitter Sentimental Analysis system and its interaction with the outside processes. Twitter Sentimental Analysis is the central system that serves as the central processing unit and receives and processes data to the supporting modules. The system is fed with search keywords, raw tweets, datasets and user metadata from Data Collection entity using query parameters like date range and number of tweets. This data is the input of the first flow in the system. This data is then sent to Data Preprocessing in which tweets are cleaned, tokenized, and normalized to be ready to analyze them. The preprocessing module sends back the processed and ready-to-review datasets to the main system.

The processed information is sent to Sentiment Analysis module, which estimates the polarity scores and attributes a sentiment label to the tweets. The sentimental-tagged database is sent to the central system and subsequently sent to the Visualization and Report module. This last step transforms the results of the analysis into graphs, summary reports and visual representations that enable users to gain insights into unfolding trends in a short period. The context diagram identifies the general interaction between key system elements, this is where data flows in and out of the system starting with data collection, passing to preprocessing, analysis and reporting but the central system remains the fundamental controller of all activities.

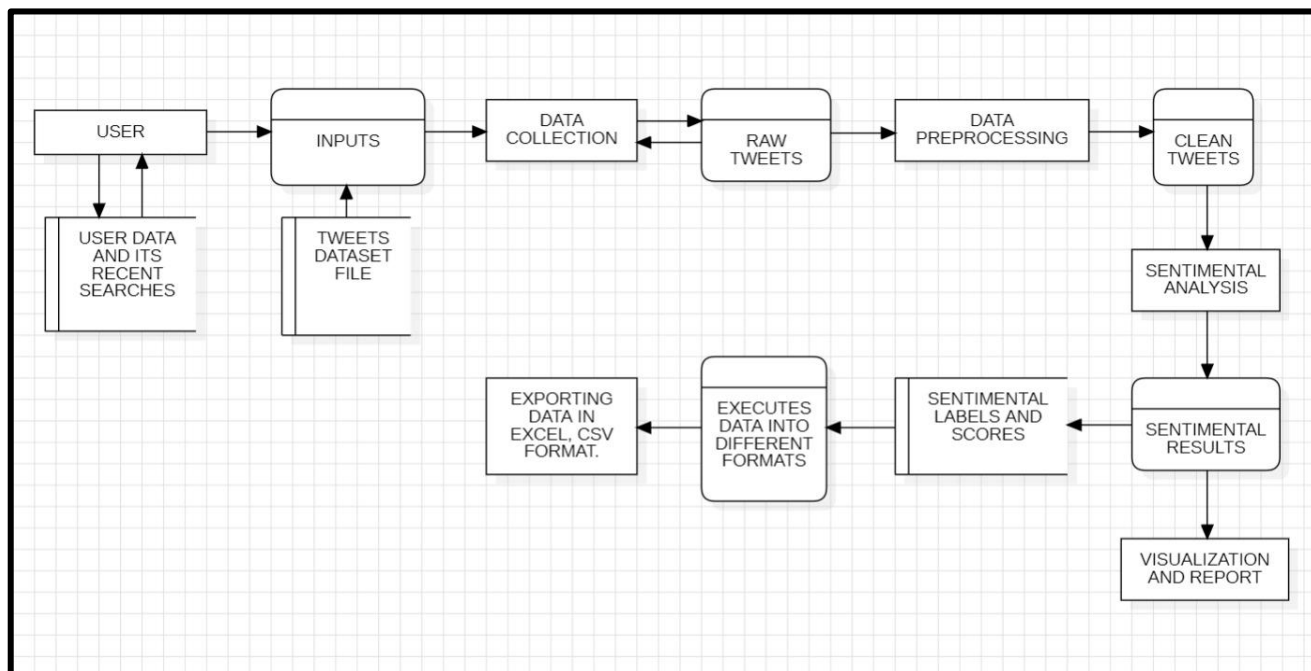
2.3. ENTITY RELATIONSHIP DIAGRAM (ERD):



The ERD is the logical structure of the TweetSense system and shows how the flow of data is from the collection of tweets to sentiment reporting. The User entity holds information for accounts and determines who is doing searches within the system. Each user is associated with several records of tweets, which contain the tweet text, language, keywords, and other metadata from the tweets. Tweets are grouped under a Dataset entity which represents a batch of collected tweets from a particular fetch operation, so that we can organize and trace data sources within the system. This structure provides the technical solution of preserving the raw tweet information while maintaining a clear link between the users and the collected data.

Before the sentiment analysis, tweets are converted to a Preprocessed Tweet entity that holds the results of cleaning, tokenization of words, and stopword removal. Separating raw and processed data helps to be more accurate, and to reuse the cleansed content without having to repeat the preprocessing. These processed tweets are then passed to the Sentiment Analysis entity and depending on the polarity scores and sentiments are recorded to indicate emotional tone. The results are summarized as a Visualization Report entity, which holds report metadata and information about charts that are used to display analytical insights. Together, these entities create a structured pipeline with capabilities to help the processing of data in a systematic manner to detect sentiment accurately along with reporting clearly and assuring scalability and traceability throughout the pipeline.

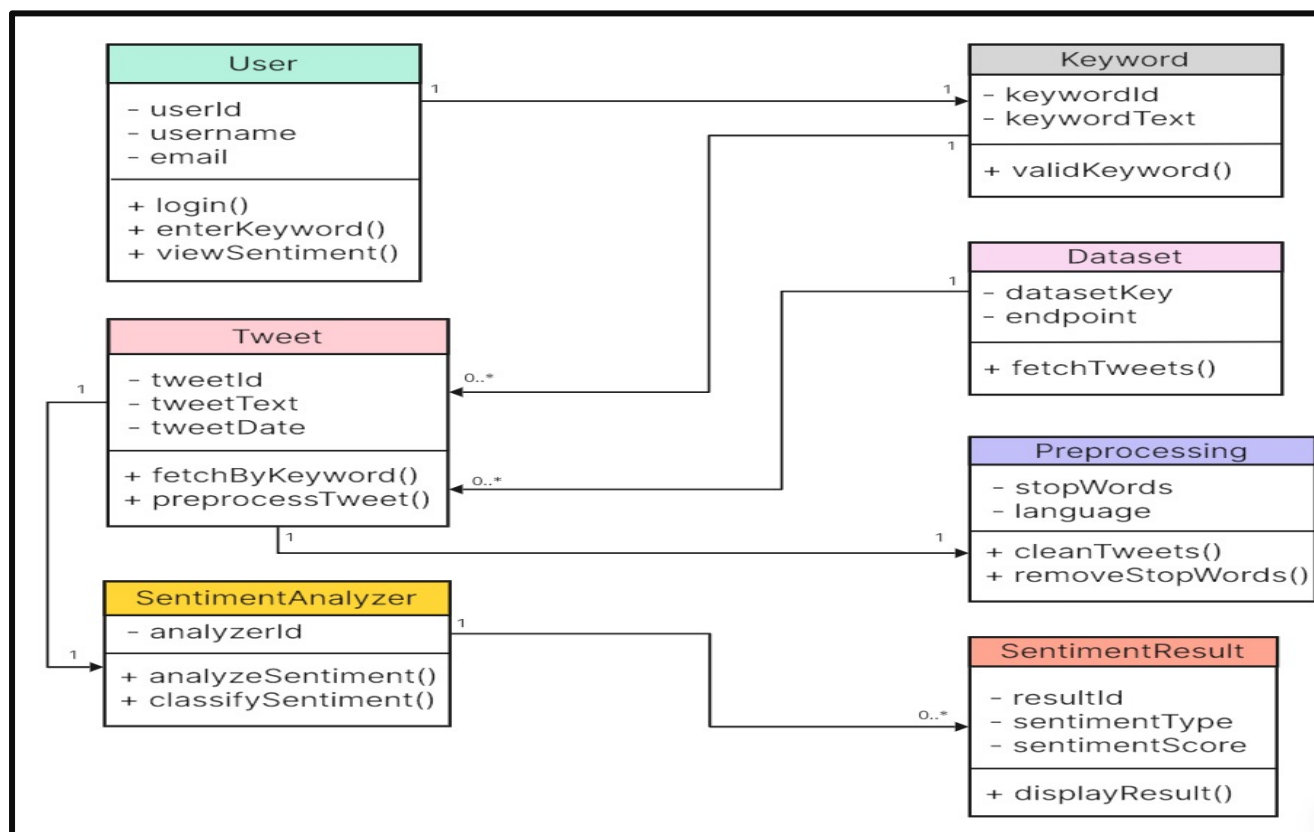
2.4. DATA FLOW DIAGRAM (DFD):



The Data Flow Diagram (DFD) - shows how data is transferred through the TweetSense system from user inputs to final reports. The process starts because the User has entered keywords or search parameters as inputs. These inputs are passed to the Data Collection process which fetches Raw Tweets from the dataset. The collected tweets are sent to the Data Processing stage in which the text is cleaned and standardised. This step is used to remove unnecessary symbols, noise, and irrelevant content and deliver Clean Tweets which guarantee data fit for analysis. The cleansed data is then fed into the Sentimental Analysis process during which each tweet is evaluated and given sentiment labels and assigned numerical scores.

The output of the analysis becomes Sentimental Results which flows to the Visualization and Report module. This process takes raw analytical output in the form of visual summaries and helps the user understand what the public thinks in a short period of time. At the same time, sentiment labels and processed data is passed on to a stage of conversion where executed data is converted into structured formats. The system has support for exporting the processed results to an Excel file and CSV file for external storage and further analysis. Overall, the DFD provides a continuous flow of data from collection, processing, analyzing, reporting, and exporting which includes the transformation of unstructured data from tweets and turning them into structured and actionable insights.

2.5. CLASS DIAGRAM:

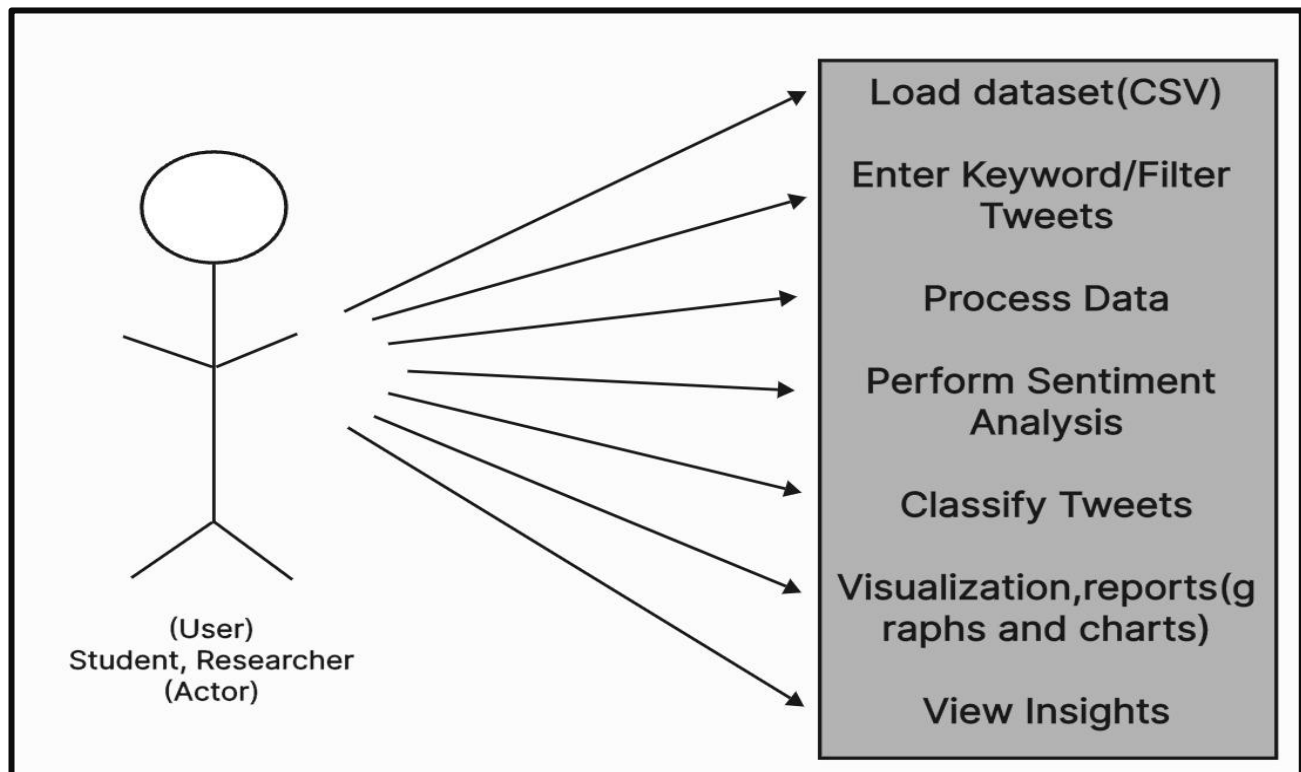


The class diagram is the object-oriented design of the Twitter Sentimental Analysis system and determines the way of interaction of the key classes in order to conduct the analysis of tweets. User class stores the user information like user ID, username, and email and offers these operations as login, key-word entry and sentiment result viewing. On entering a Keyword, a user validates the search text and links to the Dataset class that will fetch tweets to an external source. The fetched tweets are stored in the Tweet class which stores tweet id, text and date. Tweet class consists of methods of retrieving tweets based on key words and forwarding them to preprocessing. The figure indicates that there is a one-to-many association between key words and tweets, such that a single key word query could give more than one tweet.

Preprocessing class is going to cleanse the data and remove the stop words so that the text in the tweets is consistently ready to be analyzed. Tweets are cleaned and given to the SentimentAnalyzer class where emotional tone is measured through analysis and classification functions. Each of the tweets that is analysed returns a Sentiment Result object that contains the type of sentiment and the Polarity score. The relationships are used to depict the interactions of the objects in a pipeline: user input causes the tweet to be retrieved, preprocessing the data standardizes the information and the analyzer provides the structured sentiment output. Overall, the class diagram enables to emphasize

the design of modules, responsibilities, and the interaction of objects, that fight scalable and maintainable processing of sentiments that analyze.

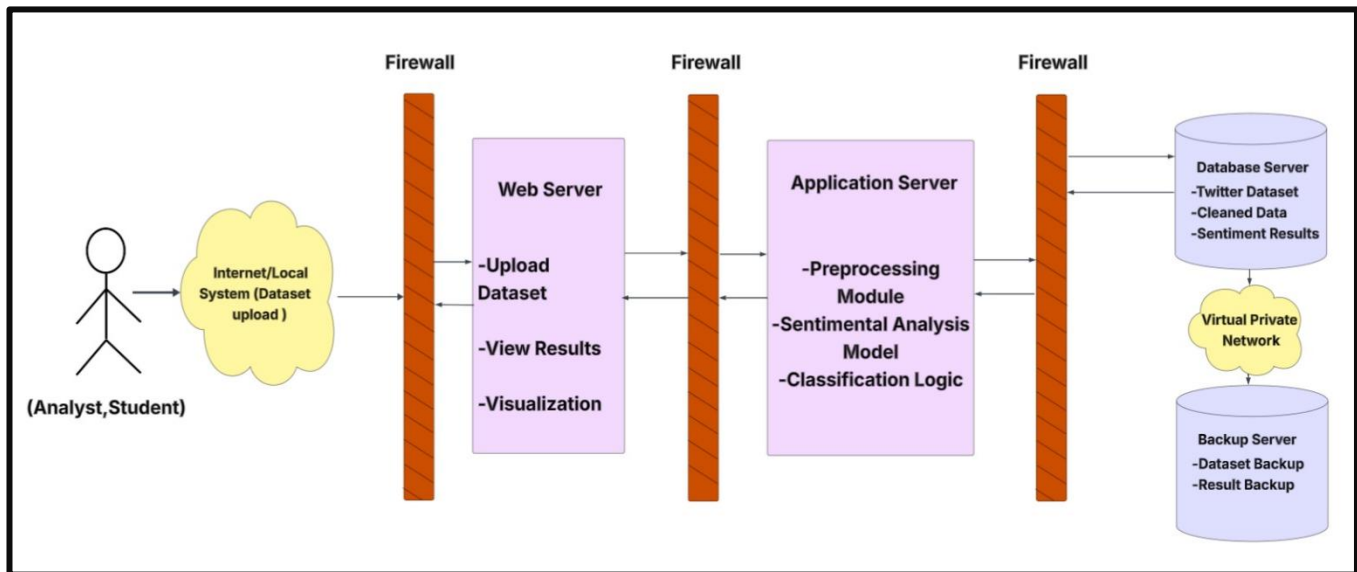
2.6. USE CASE DIAGRAM:



The use case diagram provides an understanding of the interaction between user and the Twitter Sentimental Analysis system in order to complete sentiment analysis projects. The first component is the User that can be a student or a researcher and is the first actor to make all operations in the system. This can be initiated by loading a dataset, which is in the form of CSV, or typing keywords to filter the twitters. The system then puts the chosen data through the system which gears up the data to be analyzed. When a user has processed the data, he/she invokes the sentiment analysis feature whereby tweets are assessed against sentiment categories, including positive, negative, and neutral.

After the classification, the system produces visual results in a structured form as graphs, charts, and well-structured reports. Such visualizations aid the user to interpret the emotional tendencies in the data. The last use case will enable the user to access insights and make decisions using a summarized result of sentiment. In general, in the diagram, one can observe a sequential interplay of the user governing every step of the working process, starting with the data input and analytical results, which emphasizes the usability of the system and its aim to convert the raw data of tweets into the meaningful visual results.

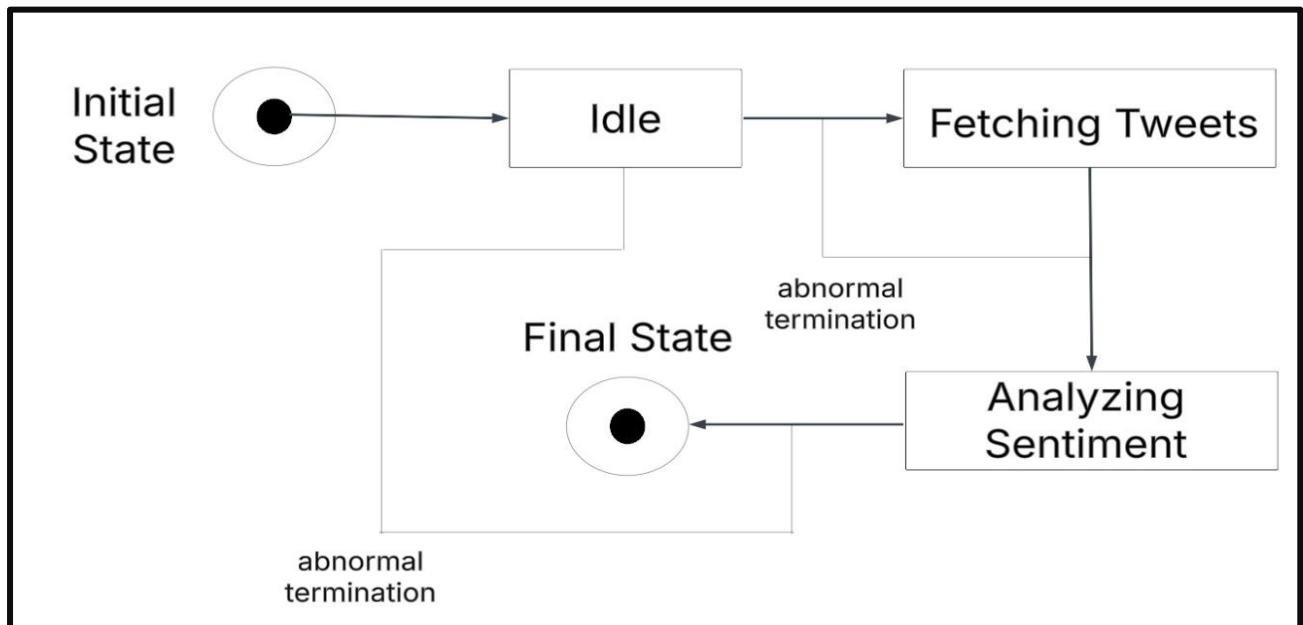
2.7. ARCHITECTURAL DIAGRAM:



The system architecture in the diagram of Twitter Sentimental Analysis platform demonstrates the interaction of various servers in order to process and secure data. The internet interface or local interface makes datasets uploadable to the user who is represented as an analyst or a student. The request is then sent through a firewall to the Web Server which receives the uploads of datasets and views the results and accesses visualization. The web-server is the presentation layer that handles interaction between the users and it does not allow direct access to the internal processing elements. The data are then safely sent to another firewall onto the Application Server, which carries out a preprocessing phase, the sentiment analysis model and the classification logic.

Results are sent to the Database Server after processing via a secure channel, where the outcomes of the tweets in the form of data sets of tweets and cleansed data, as well as sentiment data sets are stored. A Virtual Private Network (VPN) links the central database with a backup server which will ensure there is redundancy of data and restore it in case of failure. The backup server is also used to keep datasets and result copies in order to be reliable. The numerous layers of firewalls show that there was a secure architecture that segregated system components, secured sensitive information and enforced safe communication between servers. Altogether, the illustration shows a controlled and well-organized deployment infrastructure to implement scalable data processing and trusted sentiment analysis services.

2.8. STATE TRANSITION DIAGRAM:

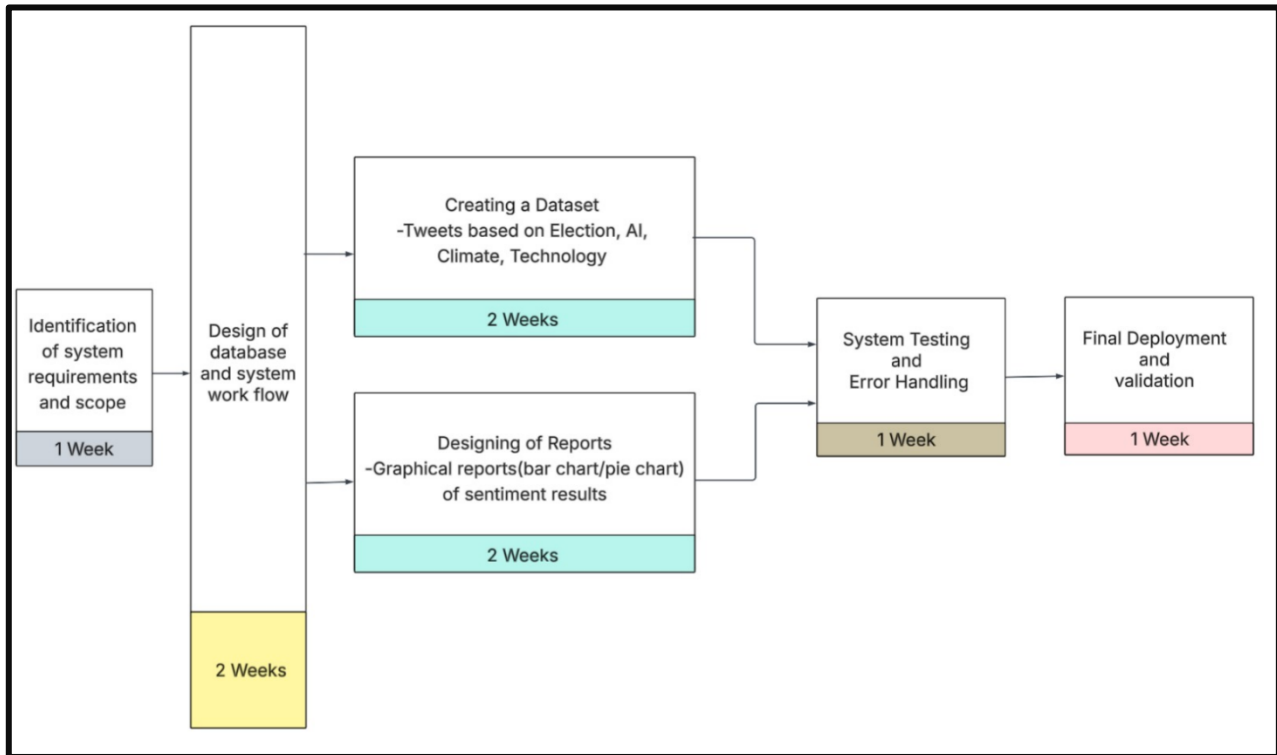


The state transition diagram is the behavior pattern of the system of Twitter Sentimental Analysis as it passes through various operational states. It starts where the Initial State is then followed by the Idle state where the system waits until a request occurs or the user makes a request to process tweets. When a request is initiated, the system goes to the Fetching Tweets state where it fetches twitters data in either the dataset or outside source. Once this is successfully retrieved, it transitions to the state of Analyzing Sentiment where tweets are then processed and categorized according to the emotional polarity. These transitions depict the usual system flow of readiness to active analysis.

System reliability is also an important issue that is pointed out in the diagram by abnormal termination paths. In case there is an error made when performing a fetch or an analysis, the system is able to stop prematurely and switch to the Final State. This will make sure that failure that might have not been anticipated will not make the system exist in an unstable state. When sentiment analysis is successfully undertaken, under normal circumstances, the system enters the final state, and the execution process is over. On balance, the diagram shows a managed lifecycle with definite phases of operations and some in-built error management to ensure that a system is stable.

➤ SYSTEM PLANNING:

3.1. ACTIVITY DIAGRAM:

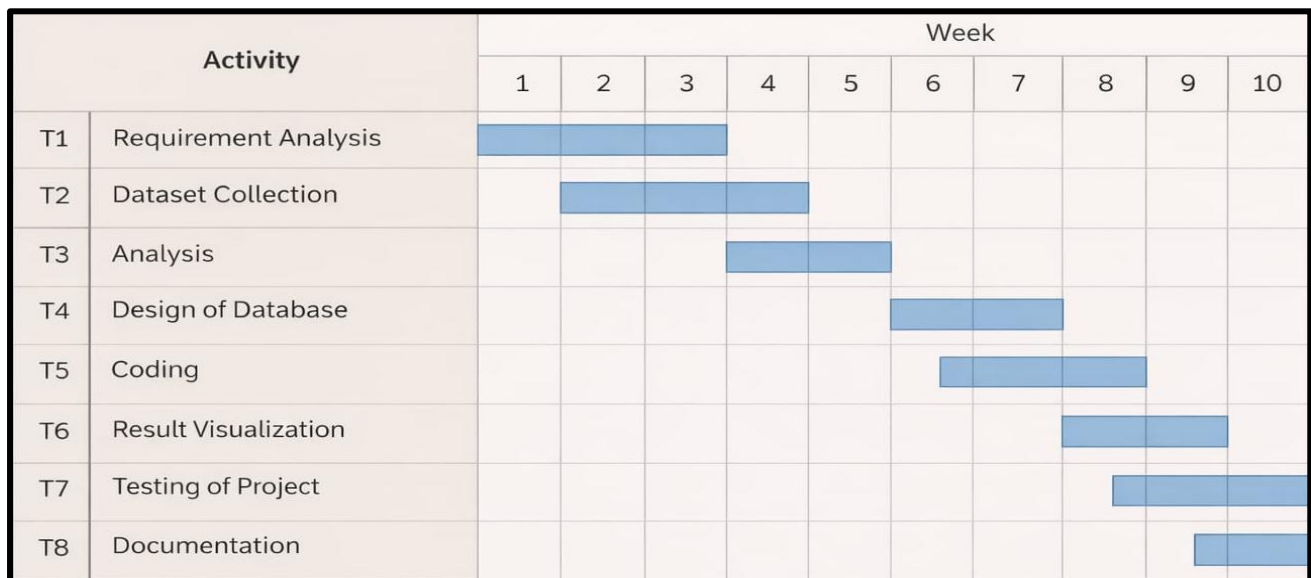


The activity diagram represents the development workflow of the Twitter Sentimental Analysis project and shows the sequence of major activities from planning to deployment. The process begins with the identification of system requirements and scope, where project goals and functional needs are defined. This is followed by the design of the database and system workflow, establishing the technical structure of the application. After the design phase, two development activities occur in parallel: creating the dataset based on selected topics such as elections, AI, climate, and technology, and designing graphical reports for sentiment visualization. These activities focus on preparing meaningful data and building reporting mechanisms that will later support analysis.

Once dataset creation and report design are completed, the workflow moves to system testing and error handling, where the application is checked for accuracy, stability, and performance. Bugs and inconsistencies are identified and corrected to ensure reliable operation. The final stage is deployment and validation, where the system is released for use and evaluated in a real environment. The diagram highlights a structured, time-bound development cycle that ensures careful planning, parallel execution of tasks, testing, and final delivery, leading to a stable and functional sentiment analysis system.

3.2.

3.3. GANTT CHART:



The Gantt chart is a system-documented development plan of the Twitter Sentimental Analysis project that takes nine weeks to complete, the way activities are planned in a reasonable order. Week 1 is the start of the project which involves the requirement analysis in which system goals, user needs and project scope have been clearly defined. This stage makes sure that development beginning is based on a sound knowledge of expectations. Weeks 2 and 3 are dedicated to database and system design during which the architecture, data structure, and workflow are to be designed such that they can be effective in enabling tweet storing and processing. Good design phase eliminates mistakes in the future and enhances reliability of a system.

Weeks 4 and 5 will be dedicated to the preparation of the data, i.e., the collection of the tweets, their filtering and assembly into usable datasets. This step is deemed significant due to the fact the quality of input data is directly connected to the accuracy of sentiments. The 6 th week is allocated to visualizing the results allowing one to easily create the charts and graphical reports displaying sentiment results. Training and testing in week 7 will be done where the performance of the system is tested and the problems are solved. Week 8 is devoted to documentation and it is necessary to record every part and process. The last week will be dedicated to the verification of the project and its validation that the system has been developed to satisfy the requirements and is conditioned to be deployed. On the whole, the chart is indicative of a strict schedule that considers planning, development, testing, and evaluation to make the project successfully finish.

➤ **SYSTEM IMPLEMENTATION & MODULE WISE EXPLANATION:**

4.1. OVERVIEW OF SYSTEM IMPLEMENTATION:

The application of system Twitter Sentimental Analysis is aimed at converting raw data of Twitter tweets into used sentiment information in the form of a series of clearly defined processing patterns. It is constructed based on the modular architecture which isolates data collection, preprocessing, analysis, storage, and visualization. This model will make sure that every part works autonomously but provides contribution to the entire work process. Its implementation is more focused on efficiency, precision and usability, meaning that the system can process considerable amounts of tweet information without affecting its performance.

The most fundamental aspect of the implementation is a pipeline that starts with acquisition of the data and culminates into graphical reporting. Tweets are received and stored in an organized database. The preprocessing step filters and standardizes the text to obtain better results on sentiment detectors. An engine based on sentiment analysis then analyzes the emotional tone, and generates labels of polarity. The findings are saved and presented in the form of charts and reports which enable users to interpret the opinion of the people in a fast manner. It is a system that uses web technologies, database management, and natural language processing software and is able to provide a reliable and scalable platform. On the whole, the implementation will be a realistic use of data analytics methods in the actual social media monitoring.

4.2. MODULE WISE EXPLANATION:

4.2.1 Module 1: Data collection from Dataset:

➤ Introduction:

So, what is this Data Collection Module all about? Basically, it takes all the data we'll need (from our Twitter side) for the sentiment analysis portion of our project. Instead of fishing around on live Twitter feeds, it is limited to a CSV file that we already have, so everything is stable and we can reproduce the results every single time. This module leads to the loading of the dataset, filtering of tweets according to the keywords that you type and storing the selected tweets in a clean form that will be used in the next step. Think of it like the first step in the entire analysis pipeline – ensuring we are only getting the useful data forward.

➤ Objectives:

We've distilled the primary objectives to these things:

- Recognize tweets from our data set at a fast rate.
- Filter them by keywords that you enter.
- Select as many tweets as you like.
- Output cleaned tweet records for some analysis.
- Maintain the data feed as reliable and consistent.

➤ Need of Data Collection Module:

Well, how can you do sentiment analysis without the data to begin with? The buzzwords in tweets have a direct impact on how accurate our sentiment scores will be. That's why we've established this dedicated module - to track how we get and filter the tweets that we need. The dataset has bunch of different topic. When someone types in a word that is a keyword, we will go in the file and get just the tweets where the keyword is mentioned. If not any of that, for the analysis to keep flowing, the module simply takes some random ones. In short, the Data Collection Module helps us to ensure that we are doing our best to start each run with clean, filtered, ready-to-process chat data at the beginning of each run.

➤ Function of Data Collection Module:

The following is what the module is really doing:

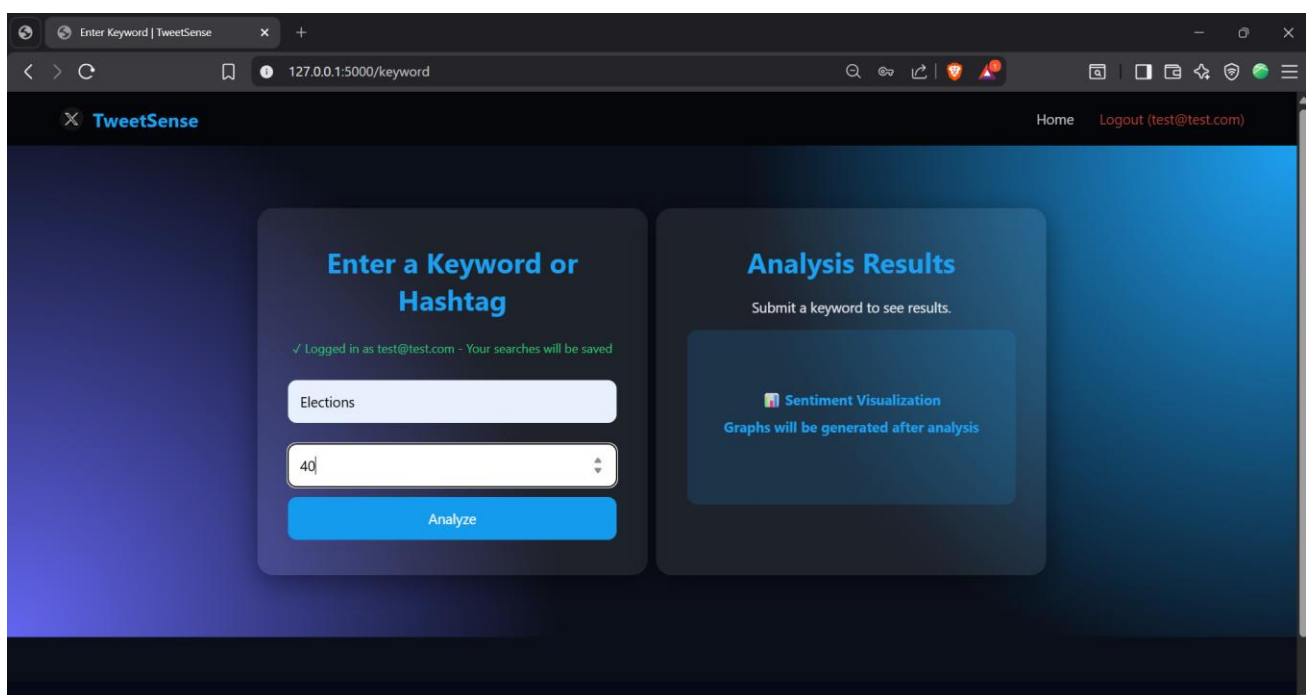
- Dataset Loading: Reading the CSV into the memory as soon as the program is initiated
- Keyword Filtering: Searches for tweets that mention the key words that you typed (case insensitive)
- Sampling: Selects the exact number of tweets that you have asked for

- **Fallback Handling:** If there is no match for a keyword it randomly samples some tweets instead
- **Data Structuring:** Modify each tweet by transforming it into a neat record containing the text of the tweet and the any metadata.

These steps ensure that only the relevant data in the correct format gets handed off the the preprocessing part of the system.

➤ Conclusion:

The Data Collection Module is the backbone of our framework of Twitter Sentiment Analysis. It ensures that each downstream process begins with regularly short, filtered and well-structured data from the tweet. In the absence of it, we wouldn't be able to rely on any decent input for preprocessing and have a shaky starting point for the end sentiment classification. As the tweets from the data set gets efficiently loaded and organized, this module allows us to perform accurate analyses, and extract real and meaningful insights about the public mood.



4.2.2. Module 2: Data Preprocessing:

➤ Introduction:

One of the most significant modules in the Twitter Sentiment Analysis system is the data preprocessing. Data on the noise in raw tweet Data includes unwanted symbols, special characters, emojis, URLs, mentions and non-uniform text formats in raw twitters. Assuming that this raw data is directly provided to the sentiment analysis model, the prediction accuracy can be reduced dramatically. The Data Preprocessing Module will involve cleaning and transforming and organizing raw data of the tweets in a structured format, which is easily analysable by the machine learning model. Simply put, preprocessing enhances the quality and consistency of data during the antecedent of sentiment classification.

➤ Objectives:

The main objectives are:

- Eliminate information that is anxious and irrelevant.
- Improve text consistency.
- Reduce data noise.
- Turn text carbon dioxide into a machine readable format.
- Increase the accuracy of sentiment analysis.

➤ Need of Data Preprocessing Module:

Twitter is a super informal and messy. People drop a lot of shorthand, emojis, hashtags, @mentions, URLs, and people are sometimes just repeating letters. All that stuff breaks sentiment models because it misrepresents what the actual meaning of the tweet is.

For example:

- Raw tweet: "I loooove this product!!! #awesome <http://link.com>"
- After preprocessing: "love product awesome"

The cleaned up version is much easier for the model to read and label. It is true that if we

➤ due not enough preprocessing, the model might do a mistake while reading or classifying the emotional vibe. So yeah, preprocessing is immensely necessary to be able to get a reliable sentiment detection.

➤ Functions of Module:

A lot of important operations are done with the Data Preprocessing Module to get tweet data ready for analysis: Hey, hey-total a Twitter story.

- Data cleaning: Addressing URLs and mentions and hashtag symbols and punctuation and all those weird special characters.
- Text normalization: I compel all the text, and squeeze all the extra spaces.

- Tokenization: I have broken the yaps down into single words so that I can work on them.
- Stopword removal: I remove the common words which don't bring significant meaning.
- Character handling over and over again: I make all those over-the-top words more normal again.
- Stemming and lemmatization I reduce the words in words so I reduce them down to root words.
- Emoji handling: I do not handle emoji or convert the emoji into its text meaning.
- Negation handling: Important negation words are kept by me so as to maintain the correct sentiment.
- With all these steps, the tweet data looks out clean, consistent, serious and actually useful for analysis.

➤ Conclusion:

Data Preprocessing Module can basically be considered as the backbone of our Twitter Sentiment Analysis system. Since tweets are so informal and completely unstructured this step ensures that the data gets cleaned, consistent, and actually makes sense before we stuff it into the sentiment classifier. If we skip the preprocessing the sentiment predictions are way off and unreliable. So this module is absolutely important for having accurate analysis and good visualization.

4.2.3. Module 3: Sentimental Analysis:

➤ Introduction:

The Twitter Sentiment Analysis system is mainly based on the Sentiment Analysis Module. This module has a role of assessing the emotional tone of the tweets, and categorizing them based on their sentiment; positive, negative or neutral. The sentiment analysis component then works on the cleaned and standardised text of the tweets in order to establish the level of polarity of the refined data. This module converts textual data to measurable sentiment scores, which allows the system to know what the population thinks or feels. It is essential in transforming cleaned tweet information into valuable analytical information.

➤ Objectives:

- Specific major goals of this module are as follows:
- Analyze cleaned tweet text
- Assign the scores of the sentiment polarities
- Classify tweets using positive or negative or neutral.
- Generate structured results in terms of sentiment
- Support proper and accurate visualizations and reporting
- Sentiment Analysis Module Requirement.

➤ Need of Sentimental Analysis module:

To comprehend the opinion of the masses, it is not enough to take a collection of tweets. The system should extract the emotional meaning between the text. Tweets are the statements of opinion, reaction, and even feelings, so sentiment analysis enables to define whether the general tone is positive, negative, or neutral.

For example:

- Tweet: "I love this new feature"
Sentiment: Positive
- Tweet: "This update is terrible"
Sentiment: Negative

The system would not be able to save text data without knowing its emotional context without this module. Sentiment Analysis Module allows the system to transform textual messages into anchored sentiment information that will be quantifiable and visualized.

➤ Functions of Sentiment Analysis Module:

The Sentiment Analysis Module has a number of major functions:

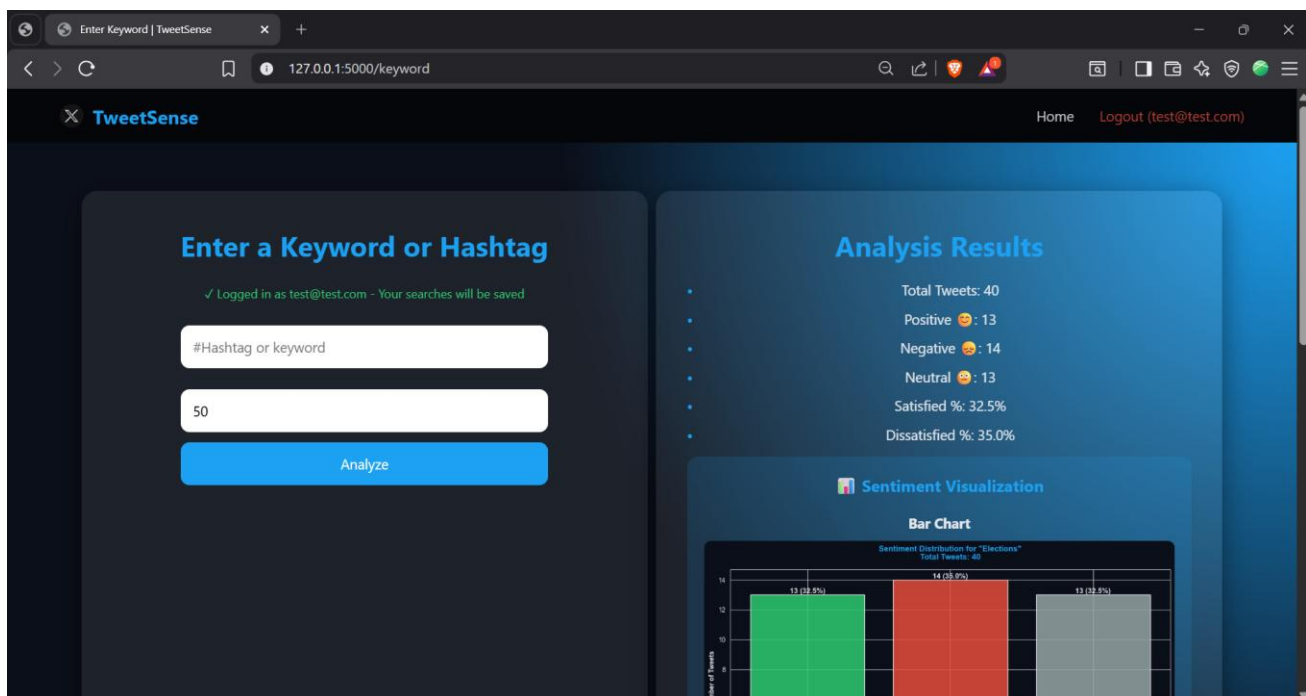
Polarity Scoring: computes numerical scores of sentiment of individual tweets.

- Classification: To assign the label: Positive, Negative, Neutral
- Threshold Evaluation-Uses predefined score values to judge the category of sentiment.

- Result Structuring Store sentiment label and polarity score.
- Statistical Aggregation: Sums the number of positive, negative and neutral tweets.
- Percentage Calculation: Percentages of satisfaction and dissatisfaction.
- These functionalities enable the system to transform processed text in tweets into insights that can be turned into action and shown in charts and reports.

➤ Conclusion:

The Sentiment Analysis Module is the smart part in the Twitter Sentiment Analysis system. It makes sense out of the cleaned tweet data and classifies them into meaningful sentiment labels with regard to polarity testing. This module provides the possibility to see the visualization and trend analysis properly by turning the textual representations into organized emotional information. In the absence of sentiment analysis, the system would lack effective ability to gauge the opinion of the population. Hence, the module is necessary in converting raw tweet content into useful results of analysis.



4.2.4. Module 4: Visualization & Reporting:

➤ Introduction:

The Visualization & Reporting Module will be in charge of transforming the obtained sentiment analysis results into the meaningful visual outputs and the organized reports. Once the sentiment classification process is over, the system produces graphical representation and summary reports to enable the users to interpret the public opinion in an easy manner. This module functions as a presentation layer of the system. It converts numerical sentiment data into charts, word clouds, and key performance indicators (KPI) and exportable documents. The module facilitates comprehension of analytical data that is complex through the presentation of results in a visual manner to both the technical user and the business stakeholders.

➤ Objectives:

The major goals of this module are:

- Prepare managed datasets for visualization.
- Analyze and display results with charts and graphical representations.
- Construct summary reports and KPI measures.
- Permit export of results in several formats.
- Offer easy to use dashboards and outputs.

➤ Need of Visualization and Reporting Module:

Raw sentiment results that are being stored in the database cannot be read directly. The visual summaries and structured reports are needed by users to comprehend trends and patterns in a short period of time.

For example:

- Instead of showing:

Positive = 120

Negative = 45

Neutral = 35

- The system displays:

Bar charts

Pie charts

Word clouds

Summary percentages

This makes the interpretation easier, and facilitates the decision-making process. In the absence of visualization, the system would only give numerical data, which may not have any significance to the stakeholders.

➤ Functions of Visualization and Reporting Module:

The module undertakes a number of major functions:

The fourth step is to prepare Visualization Datasets.

1. Prepare Visualized datasets

Laws Gets sentiment results of the Sentiment Results Repository and formats structured datasets to generate charts. It sums up measures like total number of tweets, numbers of sentiments and percentages.

2. Produce Charts and Word Clouds

Produces graphical reports, like bar charts, pie charts, trend graphs and word clouds. These visual items show the distribution of sentiments and the commonly used words.

3. Build Summary Reports & KPIs

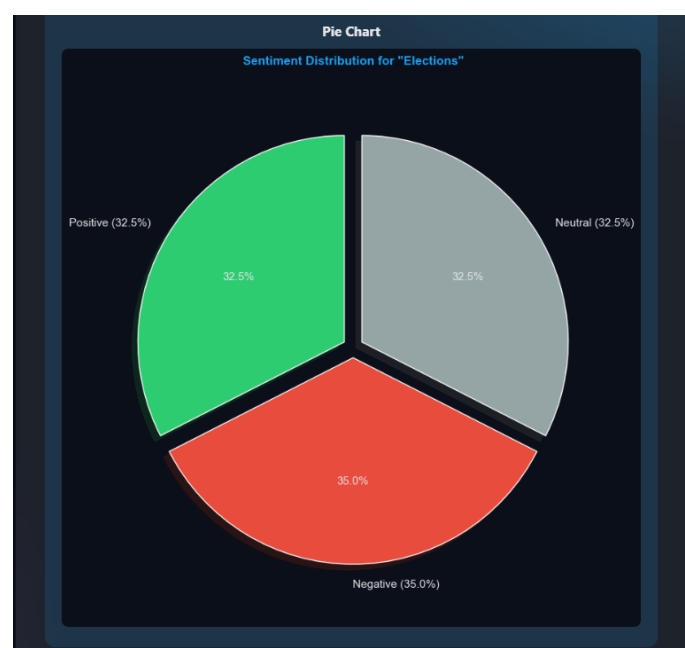
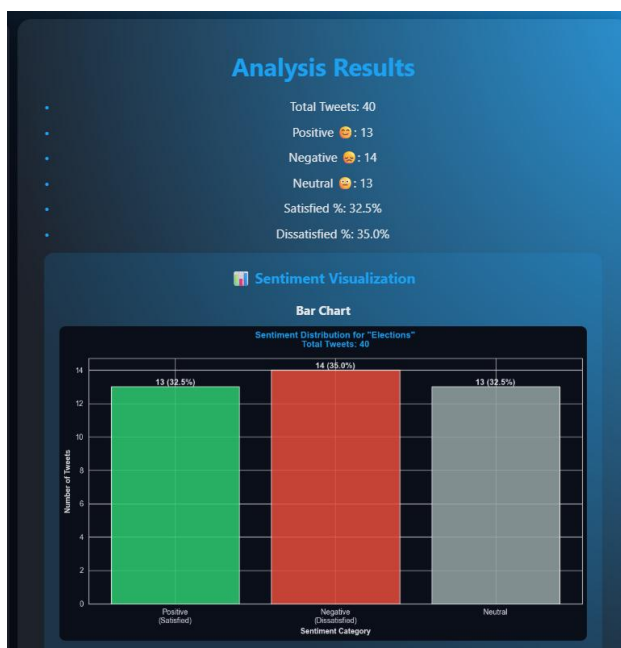
Produces executive summaries and KPIs, including percent of satisfaction and, conversely, rate of dissatisfaction. Reports can be made in the form of documents so that they can be reviewed by the management or stakeholders.

4. Export & Deliver Outputs

Gives options to users to save the analysis results in different formats like CSV, excel, PNG, or JSON. Export options through file type, naming convention and location can be set. Tracing of reporting activities are also recorded.

➤ Conclusion:

The Visualization & Reporting Module completes the sentiment analysis pipeline as it is responsible for transforming processed data into comprehensible and due insights. It enables users and stakeholders to have access to the results of sentiment through charts, reports, and through the option of exports. This module improves usability because it provides data in both graphical and structural formats, which facilitate successful decision-making. In the absence of this module, the results of the analysis would be in numbers and hard to interpret.



➤ **SYSTEM TESTING:**

In the Twitter Sentiment Analysis system, a structured approach in testing was followed to ensure the correctness, reliability and performance of the complete system. Testing was conducted after integration of all the modules that include Data Collection, Data Preprocessing, Sentiment Analysis, Database Management and Visualization. The following approaches for testing were used:

1. Black Box Testing:

Black box testing was applied to validate the functionality of systems without regard to the internal system code structure. The system was run against inputs like keywords and tweet numbers, and checking if the outputs (classified sentiments, charts, and reports) were as expected.

2. Integration Testing:

Integration testing has been carried out to assure the smooth data flow between modules:

- Data Collection - Transforming Data Preprocessing
- Preprocessing - Sentiment Analysis
- Sentiment Analysis - The Visualization
- Visualization – Export

This ensured that each module communicated in the proper manner with the others.

3. Functional Testing:

Functional testing was conducted to check if all the features worked as per the specified requirements. Each functional requirement (FR) was tested out individually.

4. Validation Testing:

Validation of data: Validation testing ensured that inputs from the user such as empty keyword or invalid number of tweets or invalid login credentials were treated appropriately by the system.

5. White Box Testing:

Checking the internal logic and the structure of the code.

With Respect to the Project: Here,

- Checking `clean_text()` logic
- Checking sentiment score computation.
- Checking database insert queries.
- Checking loop conditions of processing datasets.

6. Gray Box Testing:

Integrated Black Box + White Box.

Example:

- Knowing of preprocessing logic and testing edge cases
- Testing for thresholds of sentiments (i.e. compound > 0.05 - Positive)

7. Stress Testing:

Trial of the system at extreme conditions.

Example:

- There are repetitive processing of maximum number of tweets.
- Posting extremely huge data set.

8. Reliability Testing:

Checking if system will run continuously/non-stop.

Example:

- Conducting several searches at a time.
- Repeated export operations

9. Recovery Testing:

Recovery of the testing systems on failure.

Example:

- Database connection loss.
- Server restart
- Interrupted file upload

10. Compatibility Testing:

Testing system on other environments.

Example:

- Chrome browser
- Edge browser
- Different Resolutions of Screen

5.1. TEST CASE TABLE:

TC ID	DESCRIPTION	MODULE	INPUT	EXPECTED OUTPUT	RESULTS
TC – 01	Upload dataset	Data Collection	CSV File	Dataset stored successfully	PASS
TC – 02	Preprocessed tweet	Preprocessing	Raw tweet with URL	Cleaned tweet generated	PASS
TC – 03	Stopword removal	Preprocessing	Tweet with stopwords	Stopwords removed	PASS
TC – 04	Tokenization	Preprocessing	Clean text	Tokens generated	PASS
TC – 05	Sentiment classify	Sentiment Analysis	Cleaned tweet	Positive/Negative/Neutral	PASS
TC – 06	Bulk analysis	Sentiment Analysis	Multiple tweets	All tweets classified	PASS
TC – 07	Generate charts	Visualization	Sentiment data	Bar & Pie charts	PASS
TC – 08	Export report	Reporting	CSV/PDF option	File downloaded	PASS
TC – 09	Invalid file	Validation	Unsupported file	Error shown	PASS
TC – 10	Empty input	Validation	Blank keyword	Validation message	PASS

➤ **COST ESTIMATION REPORT:**

6.1. PROJECT SIZE SUMMARY:

CATEGORY	VALUE
Estimated KLOC	1.5 KLOC
Total Source Code	~1513 Lines
Frontend Files	HTML, CSS, Templates
Backend Files	Python (Flask) Modules
Database Tables	Users, Searches, Results
Visualization Modules	Bar Chart, Pie Chart
Python Libraries	NLTK, Pandas, Matplotlib
Total Modules	4 Major Modules

6.2. TECHNOLOGY STACK:

LAYER	TECHNOLOGY
Frontend	HTML, CSS, Jinja Templates
Backend	Python (Flask Framework)
Database	MySQL (WAMP Server)
NLP Processing	NLTK Library
Data Processing	Pandas
Visualization	Matplotlib
Other Tools	PyMySQL, SQLAlchemy

6.3. COCOMO (CONSTRUCTIVE COST MODEL) ESTIMATION:

Formulas:

- Effort (Person Months) = $a \times (\text{KLOC})^b$
- Development Time (Months) = $c \times (\text{Effort})^d$
- Team Size = Effort / Development Time

CONSTANT	VALUE
a	2.4
b	1.05
c	2.5
d	0.38

➤ **CALCULATION:**

PARAMETERS	FORMULA	RESULT
KLOC	1,513 / 1,000	1.513 KLOC
EFFORT	$2.4 \times (1.513)^{1.05}$	Effort $\approx$ 3.71 Person Months
DEVELOPMENT TIME	$2.5 \times (3.71)^{0.38}$	Development Time $\approx$ 4.11 Months
TEAM SIZE	3.71 / 4.11	Team Size $\approx$ 1 Developer

6.4. FUNCTION POINT ANALYSIS (FPA):

➤ **STANDARD IFPUG WEIGHTS:**

FUNCTION TYPE	AVERAGE WEIGHT
External Inputs (EI)	4
External Outputs (EO)	5
External Inquiries (EQ)	4
Internal Logical Files (ILF)	10
External Interface Files (EIF)	7

➤ **FUNCTION POINT CALCULATION:**

TYPE	COUNT	COMPLEXITY WEIGHT	TOTAL FP
External Inputs (EI)	5	Avg = 4	20
External Outputs (EO)	4	Avg = 5	20
External Inquiries (EQ)	3	Avg = 4	12
Internal Logical Files (ILF)	3	Avg = 10	30
External Interface Files (EIF)	2	Avg = 7	14

➤ **UNADJUSTED FUNCTION POINTS (UFP):**

FORMULA:

$$\text{UFP} = \sum (\text{Count} \times \text{Complexity Weight})$$

$$\text{UFP} = 96$$

➤ **VALUE ADJUSTMENT FACTOR (VAF):**

FORMULA:

$$\text{VAF} = 0.65 + (0.01 \times \sum F_i)$$

$$\text{VAF} = 0.65 + (0.01 \times 40)$$

$$\text{VAF} = 0.65 + 0.40$$

$$\text{VAF} = 1.05$$

➤ **FINAL FUNCTION POINTS:**

$$\text{FP} = \text{UFP} \times \text{VAF}$$

$$\text{FP} = 96 \times 1.05$$

$$\text{Final FP} = \mathbf{100.8}$$

➤ **FPA BASED COST ESTIMATION:**

$$\text{Cost} = \text{Effort} \times \text{Cost per Person Month}$$

$$\text{Cost} = 3.71 \times 25,000$$

$$\text{Estimated Cost} = \mathbf{₹92,750}$$

6.5. COST ESTIMATION SUMMARY:

PARAMETER	REMARKS	VALUE / RESULT
Project Size	~1513 lines of code	1.513 KLOC
Effort (Person Months)	Calculated using COCOMO model	3.71 PM
Development Time	Estimated project duration	~4.11 Months
Team Size	Effort $\div$ Development Time	1 Developer
Unadjusted Function Points (UFP)	From Function Point Analysis	96
Value Adjustment Factor (VAF)	Based on $\Sigma Fi = 40$	1.05
Final Function Points (FP)	$UFP \times VAF$	100.8
Cost per Person Month	Estimated developer monthly cost	₹25,000
Estimated Development Cost	Effort $\times$ Cost per Person Month	₹92,750

➤ CONCLUSION:

Twitter Sentiment Analysis System that was created in the given project is a complex analytical tool that is created to process, analyze and interpret the opinions of the population expressed on the social media. The system translates unstructured data of tweets to formal and valuable information, which uncovers the trend of public sentiment. The system will save a lot of time, accuracy and efficiency otherwise used in manual analysis methods that are considered very time consuming, inaccurate and inefficient in collecting, preprocessing, sentiment classification, visualization and reporting data.

Hand munging of tweets in large quantities is an energy-intensive process that is susceptible to bias and inconsistency. The challenges are conquered by the implemented system as it combines Natural Language Processing methods with database management and visualization tools. The system enhances the accuracy of sentiment detection through the process of preprocessing which includes cleaning, normalization, tokenization, and removal of stopwords. To ascertain the sentiment of a set of tweets in a fast, reliable, and consistent manner, the sentiment analysis module categorizes tweets into positive, negative, and neutral through a mechanism of polarity scoring to provide a reliable, fast, and less expensive interpretation of opinions.

The project was created according to the Software Development Life Cycle (SDLC) consisting of the systematical planning, design, implementation, testing and deployment. The modules were developed separately and then interpreting to provide the functionality. The reliability and performance of the system was tested and confirmed by functional, integration as well as validation testing. Modular design also provides maintainability and scalability so that future upgrades would not interfere with the current features.

One of the strengths of the system is that it is able to show analytical results in a clear visual format using bar charts, pie charts, and word clouds. The reporting and export capabilities also contribute to the usability because it allows creating summaries in various formats. This renders the system useful in academic studies, business intelligence, marketing studies, and decision support.

To sum up, the Twitter Sentiment Analysis System has managed to illustrate how Natural Language Processing, database management, and web technologies can be practically applied to resolve analytical issues in the real world. It enhances efficiency, manual workload, decision-making, and gives insightful results about the opinion of people. The project meets its goals and creates a substantial underpinning of further betterments and higher application levels of sentiment analysis.

➤ **PLANS OF THE TWITTER SENTIMENT ANALYSIS SYSTEM IN FUTURE:**

Currently, the Twitter Sentiment Analysis System is useful in collecting tweets within a dataset, processing the information, sentiment classification, and creating visual reports. Nevertheless, the system can be improved in a number of ways and become more intelligent, scaled, and feature-filled.

A] Real-Time_Tweet_Collection:

At the present, the system is compatible with a fixed set of data. It can be expanded in the future to retrieve official APIs of real-time tweets. This would enable real-time tracking of what people think about trending topics, events or brands thus making the system more active and practically applicable.

B] State-of-the-Art_machine_learning_models:

The lexicon based or basic sentiment classification method is used in the current system. Additional progress may be made in future with more sophisticated models of machine learning or deep learning like LSTM, BERT, or transformer-based systems. These models would become more accurate at least in terms of the detection of sarcasm, context-based sentiment, and complex language patterns.

C] Sentiment_Analysis_in_Multilingualism:

Presently, the system primarily handles English tweets. It can be improved to accept more than one language, enabling one to analyze the opinions of the global population. This would render the system more flexible and able to apply to international research and business cases.

D] Expansion_of_mobile_and_Web_Applications:

The system can be scaled to a full-fledged web portal or mobile app where one is able to log in, trending keywords, and see real-time dashboards. This would make it more accessible and enable users to track sentiment intelligence everywhere.

E] Automated_Notifications_and_Alerts:

Automated notifications can be added to the system in the future. As an illustration, should negative sentiment reach a specific threshold, the system might raise the alarm of the administrators or stakeholders. This would be of great help especially in monitoring the brands and crisis management.

F]_Business_Intelligence_Tools_Integration:

The system could be linked with outside Business Intelligence (BI) applications like Power BI or Tableau. This would enable the higher order of dashboard development and style of enterprise reporting among corporate users.

G]_Improved_Visualization_capabilities:

It can be further enhanced with interactive dashboards, time-series analysis graphs, and heat maps and comparative sentiment analysis across the topics in the future. These would give a better analytical information and user experience.

H]_Cloud-Based_Deployment:

Implementation of the system on cloud platforms would facilitate scaling of the system, data storage and ease of access. The deployment to the cloud would also enable the management of larger dataset and users at the same time.

I]_Better_Access_Control_and_Security:

Its future improvements could contain role-based access control, encrypted data storage, secure mechanism of logging-in as well as logging activities. This would enhance the security of the systems and safeguard the users information.

J]_Predictive_Trend_and_Sentiment_Forecasting:

The system can be further expanded into predictive analytics. Through the development of the past sentiment patterns, the system would know the future trends, or how the people may respond to any given event or product in the future.

➤ **CONCLUSION OF FUTURE SCOPE:**

To conclude, although the existing Twitter Sentiment Analysis System can accomplish its goals, it can be improved in the future. With the introduction of real time data processing, state-of-the-art machine learning algorithms, multilingual support, cloud deployment and intelligent alert systems, the system will become a powerful, enterprise-level sentiment intelligence platform. Not only would these innovations increase its analytical potential, but it would also increase its usability in research, marketing, governance and strategic decision making.

➤ REFERENCES:

- Diagrams and Designs: Star UML, Draw.io, Chatgpt
- Some Points are extracted from: Perplexity, Claude and Random AI Models
- Flask Documentation, <https://flask.palletsprojects.com>
- Kaggle, *Datasets and Data Science Resources*, <https://www.kaggle.com>
- Database Management: **MySQL (WAMP Server)**